

TITLE

**ELECTRICAL SAFETY
AWARENESS BOOKLET.**

BY

M.VAMSHI KUMAR (BWU/DEE/22/038),

FROM

DEPARTMENT OF ELECTRICAL ENGINEERING

Institute of Brainware University.

398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125

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AWARENESS BOOKLET.**

BY
M. VAMSHI KUMAR (BWU/DEE/22/038),

**UNDER THE SUPERVISION OF HEAD OF
DEPARTMENT**

Mr. SUBHAJIT MUKHERJEE

FROM
DEPARTMENT OF ELECTRICAL ENGINEERING

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“Safety goes Danger grows”

The abundant usage of electricity in the day-to-day activities by an individual either at home, office or place of work, be it an industry or a public place is associated with a certain degree of risk to life and property. As electrical energy strikes without warning, vigilance is necessary pre-requisite to avoid accident. Since potential danger from electricity cannot be seen, heard or smelt, it is most essential that knowledge on the extent of damage it could cause is known, so that preventive measures could be initiated in every premises and instance of such electricity use.

INTRODUCTION

Electricity is an essential part of modern life, powering our homes, schools, and workplaces—but it also poses serious risks if not handled with care. This **Electrical Safety Awareness Booklet** is designed to educate children, teenagers, and families on how to use electricity safely and prevent accidents. From common hazards like frayed wires and overloaded sockets to emergency procedures for electric shocks and lightning, this guide covers vital safety practices in an easy-to-understand format. You'll learn why certified electrical products matter, how proper earthing protects you, and what to do in dangerous situations. Whether at home, school, or outdoors, these practical tips—presented through engaging visuals and simple language—will help you recognize risks and take the right precautions. Remember, electrical safety is everyone's responsibility. By following these guidelines, you can prevent injuries, fires, and even save lives. Let's work together to stay safe and make electricity a powerful ally, not a hidden danger!

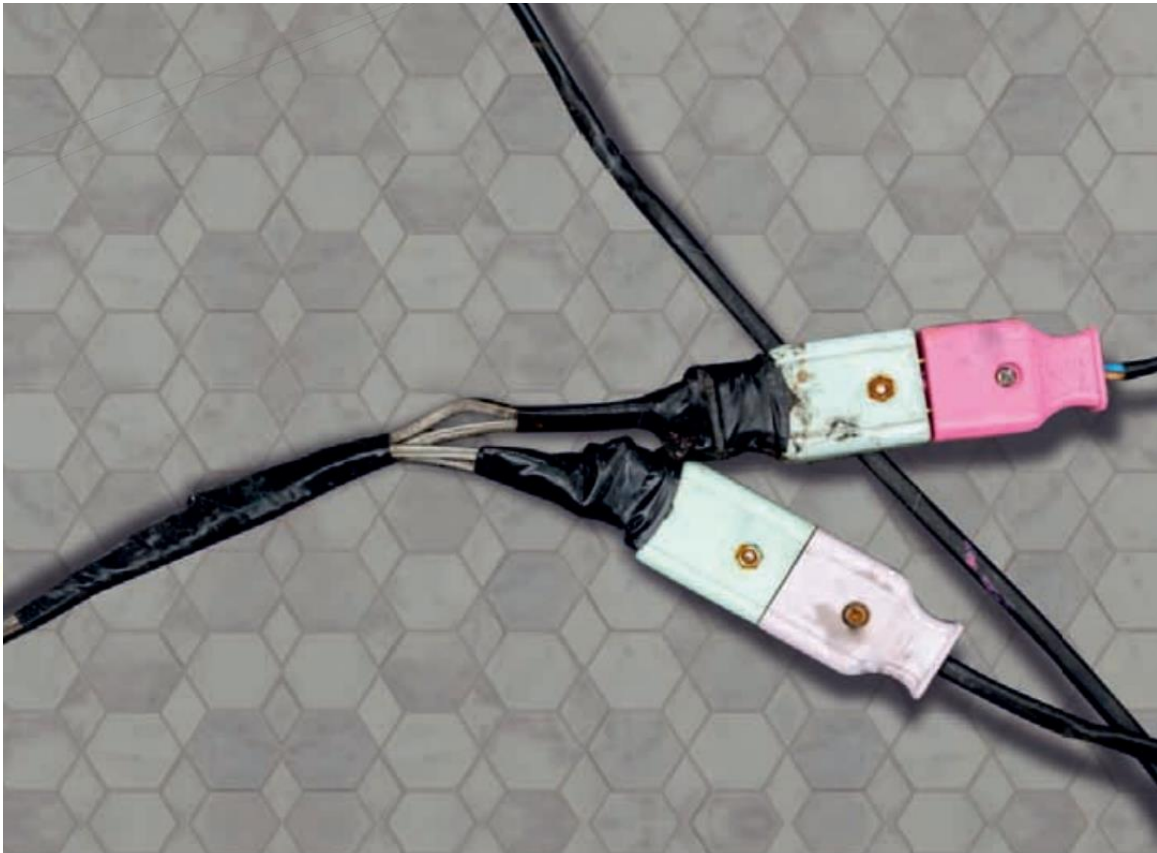
Introduction:

Why This Booklet Matters



- ✓ **Welcome to the world of safe electricity!**
- ✓ **This booklet is your guide to understanding electricity the fun and safe way.**
- ✓ **Learn through comics, visuals, and real-life tips!**

CAUSES OF ELECTRICAL ACCIDENTS & ELECTRICAL FIRE.



Nearly one third
of home
electrical fires
begin with
**ignition of wire
or cable
insulation.**



CAUSES OF ELECTRICAL ACCIDENTS & ELECTRICAL FIRE

1. Use of old tools and cables which are prone to cause earth leakage due to their inadequate insulation resistance.
2. Use of undersized wires and cables.
3. Dragging of metal plates/Street sections/equipment over cables which may result in insulation damage/ failure.
4. Repairing electrical equipment /hand tools with power ON
5. Dropping tools to live busbars accidentally.
6. Excavating buried cable route which may contain live cables.
7. Not providing route markers for buried cables.
8. Scaffolding crane boom touching overhead lines.
9. Haphazard way of taking power from distribution board without proper plug sockets or with multiple twisted joints.
- 10 Not using proper fuse wires or use of conductors as a fuse wire in emergency.
11. Use of bare fuse wire in High Rupturing Capacity (HRC) fuse carriers.
12. Entry of moisture into switch boards.
13. Assuming low voltages to be safe.
14. Use of earth/ structure as neutral or return path for glowing a bulb or operating hand tools.
15. Use of protection devices with higher ratings.
16. Wrong design of the electrical installations
17. Inadequate protection scheme
18. Insufficient clearance for cooling and maintenance of transformer
19. Lack of preventive maintenance of electrical equipment.

20. Insufficient fault isolators amounting to wrong selection of circuit breaker/load break switch.
21. Improper bus ratings
22. Not providing proper earthing to electrical equipment
23. Improper stringing of the Over Headlines without adequate clearance between phases and between phase & earth
24. No provision of anti-climbing devices on towers /poles
25. Improper termination /jointing of cable
26. No provisions of LT KioskS/LT cutouts
27. Improper location and not providing fencing the oil filled transformers
28. "Chalta-hai" attitude.

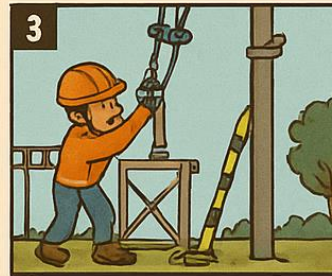
CAUSES OF ELECTRICAL ACCIDENTS & ELECTRICAL FIRE



1
Use of old tools and cables which are prone to cause earth leakage due to their



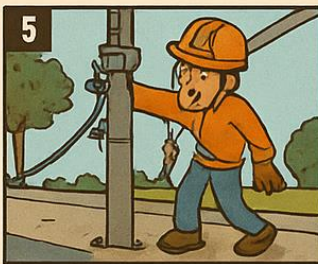
2
Use of undersized wires and cables



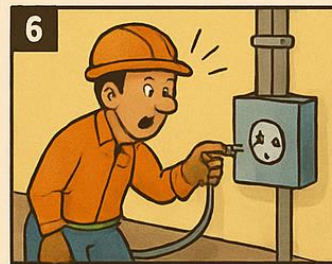
3
Dragging of metal plates/ Street sections/ equipment over cables which



4
Excavating electrical routes without markers for



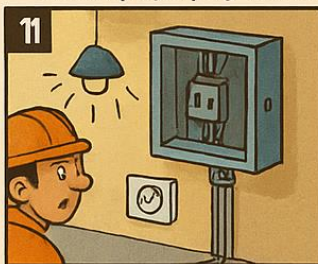
5
Haphazard way of taking power from distribution board without proper plug sockets



6
Not using proper fuse wires or using conductor as a fuse wire



7
Haphazard way of taking power from distribution



11
Entry of moisture into switchboards



14
Assuming low voltages to be safe



15
Use of bare fuse wire in High Rupturing Capacity



15
Entry of moisture into switchboards



15
Use of protection devices with higher ratings

IN CASE OF SHOCK OR FIRE 🔥



If someone is shocked:

- ❌ Don't touch them directly
- ✅ Turn off the power
- ✅ Use a wooden stick to separate them
- ✅ Call for help immediately

If there's a fire:

- ✅ Use a CO₂ extinguisher only
- ❌ Never use water!
- ✅ Raise the alarm and get help



CAUSES OF OVERLOAD & SHORT CIRCUITS

- ✓ Inferior grade materials
- ✓ Undersizing of wiring
- ✓ Improper earthing
- ✓ Insulation failure
- ✓ Loose conductors
- ✓ Poor design

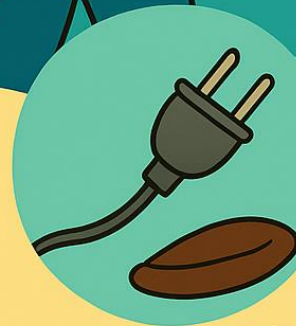


OVERLOADED
electrical circuits
are potential
FIRE
HAZARDS.

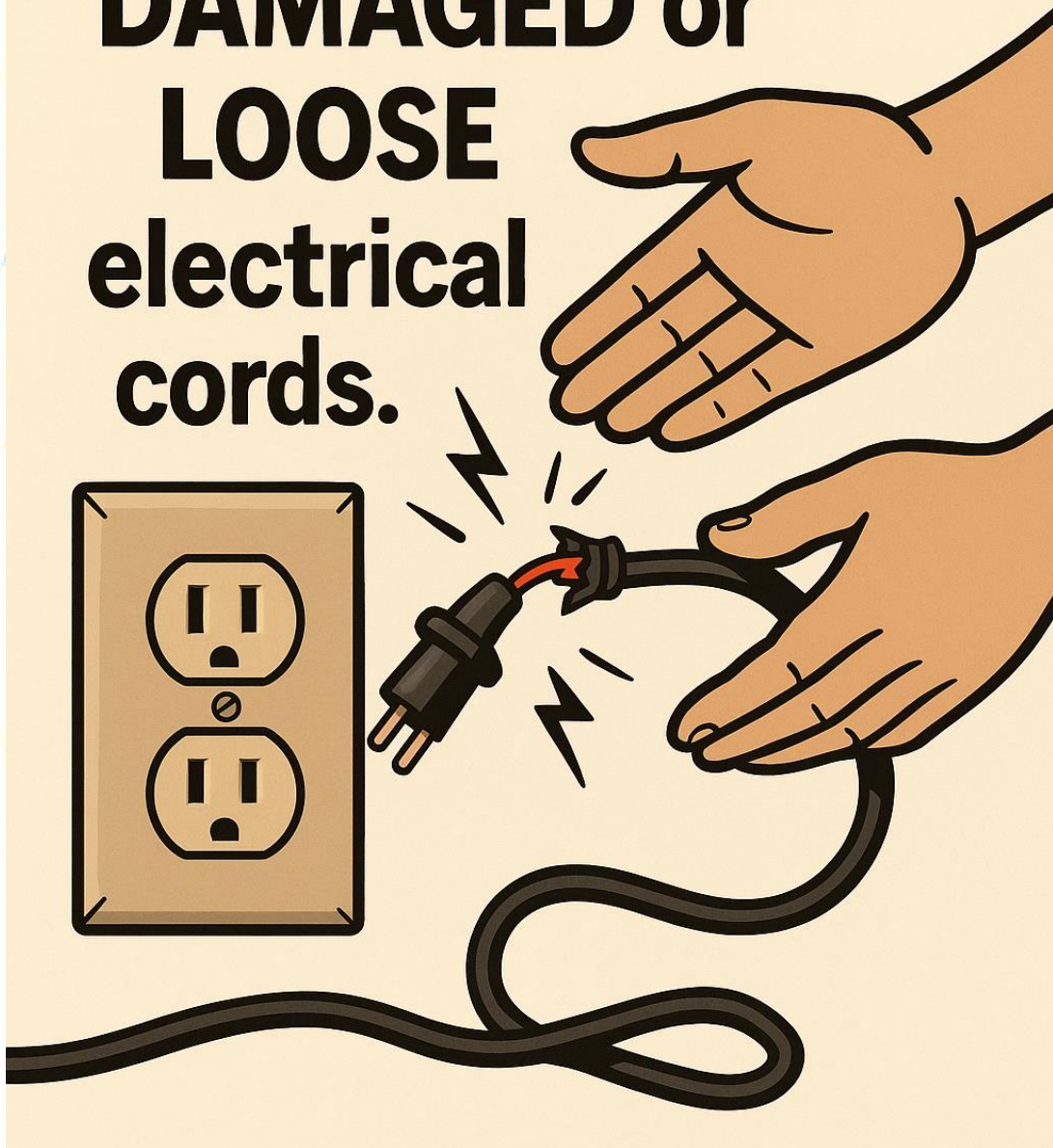
Water + Electricity = **DANGER!**



- ✓ Keep electronics away from water
- ✓ Never use plugs or cords with wet hands
- ✓ Ask an adult before using appliances near water



**Timely
REPLACE or repair
DAMAGED or
LOOSE
electrical
cords.**





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ELECTRICAL SAFETY PRECAUTIONS

A flammable Item
that is too close to a
light fixture
or bulb is the leading factor
contributing to
electrical fires.

GIVE THEM SPACE!



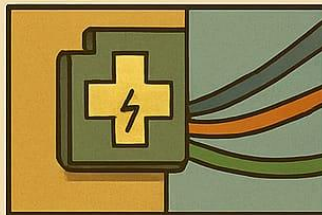
ELECTRICAL SAFETY PRECAUTIONS

- Display single line diagram, shock treatment chart, emergency contact phone numbers etc. in electrical switchgear room.
- Fix danger/ caution boards on conspicuous positions.
- Keep a good first aid kit at easily accessible place and check/ replace contents at regular intervals.
- Adopt colour coding of wires. Green colour should be used only for earthing.
- Locate switches in such a way that they are easily accessible during emergencies.
- Always follow good housekeeping.
- Ensure availability of proper protection device (ensure correct rating) in each circuit.
- Never use single pole switch or fuse in neutral.
- Never pullout plugs without switching off.
- Use only three pin plugs in single phase circuits.
- Clearly mark circuit breakers & switches to indicate the load / circuit.
- In high current and inductive circuits, switches should be provided and operated before fuses.
- Ensure overhead lines are switched off before movement of cranes in their vicinity.

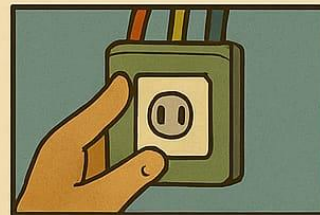
ELECTRICAL SAFETY PRECAUTIONS



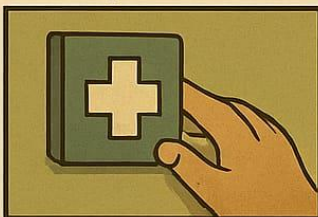
Display single line diagram, shock treatment chart, emergency contact phone numbers etc. in electrical switchgear room.



Fix danger / caution boards on conspicuous



Keep a first aid kit at easily accessible place and check / replace contents at regular



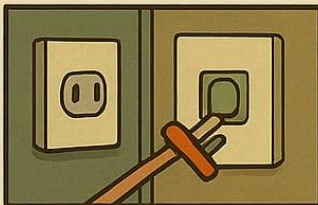
Ensure availability of proper protection device



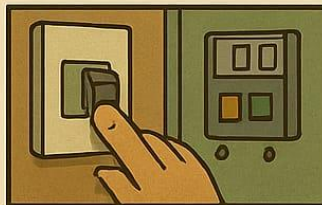
Good housekeeping



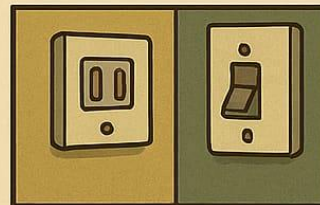
Never pull out plugs without switching off



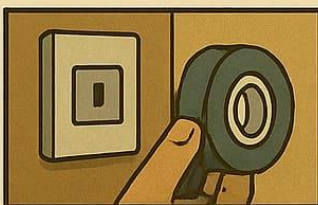
Clearly mark circuit breakers & switches to indicate the load / circuit



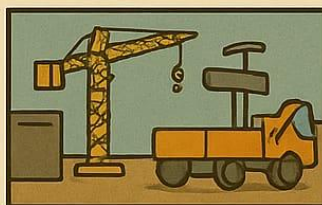
In high current and inductive circuits, switches should be provided before fuses



Ensure overhead lines are switched off before movement of cranes in their vicinity



Ensure pin operated before fuses so operated before fuses



Ensure overhead lines are switched off before movement of cranes

NEVER LEAVE ANY WIRE EXPOSED.
USE INSULATION TAPE

FOLLOW SAFE WORK PRACTICES WITH ELECTRICITY



Provide signs and barricades to warn of hazards

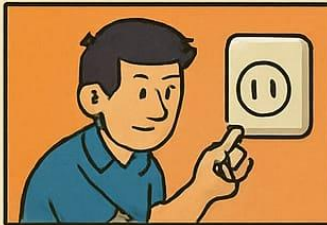
Provide signs and barricades to warn of hazards

GET ALL ELECTRICAL BEFORE THEY STU!

ELECTRICAL SAFETY IPS



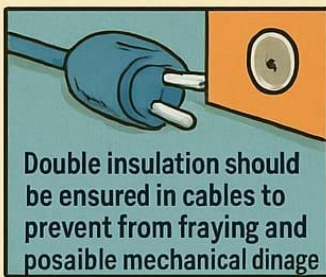
All electrical cords and appliances must have apt weather protection



Don't attach electrical cords to the building surfaces/structures



Ensure Firm mounting of receptacles in their enclosures



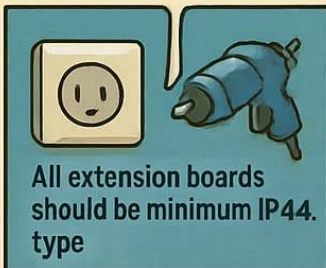
Double insulation should be ensured in cables to prevent from fraying and possible mechanical damage



Cables should be laid either underground or routed minimum 4m above the ground level



The cable should be routed above the ground with a minimum height of 5 m from the ground level where it is crossing the passage meant for vehicle movement, where underground laying is not possible



All extension boards should be minimum IP44 type

When not in use, tools should be unplugged



Ensure that the cable must be secured inside equipment by non-conducting faces so that it cannot slip

Electrical equipment that has grounded three pronged plugs should only be used



If a plug breaks off inside an outlet, do not attempt to remove by yourself

ADDITIONAL SAFETY MEASURES



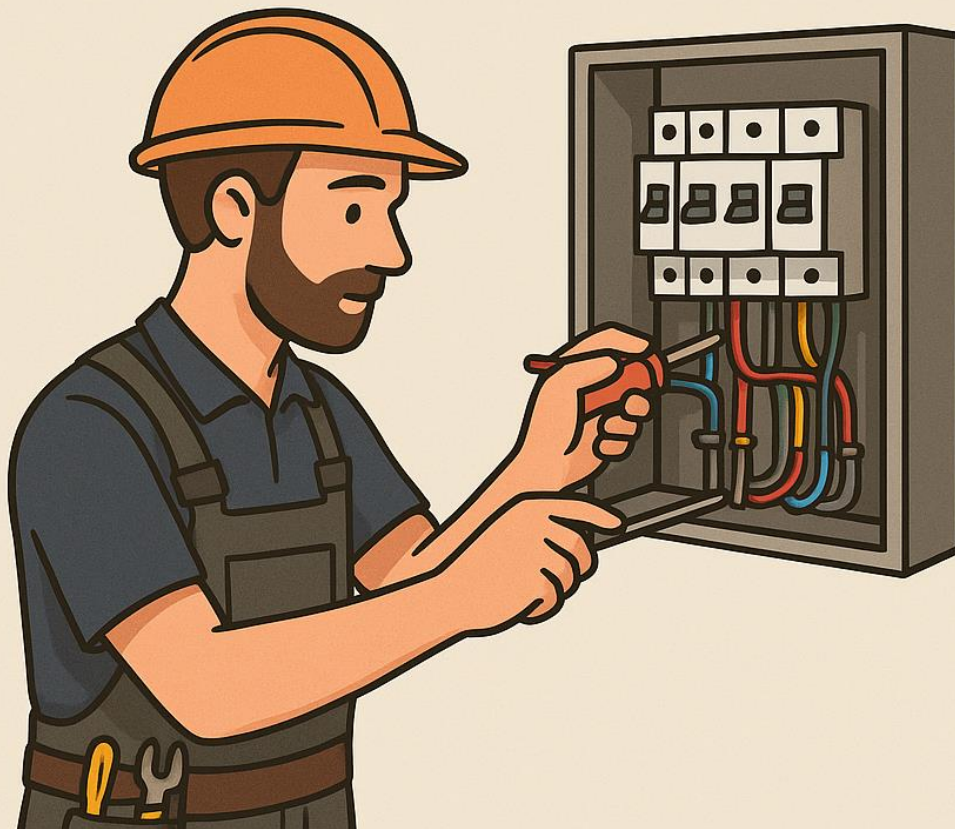
We should stay at least 10 feet away from overhead power lines

Ensure Proper illumination in all areas where electrical hazards are to be encountered. An emergency lighting system should be in place



Specialized personnel protective equipment

Get all the electrical work done only by **QUALIFIED ELECTRICIAN**



Accident brings tears **SAFETY** brings cheers

- **DO NOT** renew a blown fuse, until the cause is identified and rectified.
- **DO NOT** close any switch unless you know why it was kept open.
- **DO NOT** Touch or tamper with any electrical equipment or conductor, unless you have made sure that it is **DEAD** and **EARTHED**.
- **DO NOT** Work on live circuit without the permission of the supervisor. Make sure that All safety precautions have been taken and you are accompanied by a second person competent to render first aid.
- **DO NOT** disconnect earthing connection or by-pass safety gadgets installed on mains and apparatus.
- **DO NOT** open or close switch or fuse slowly or hesitantly. Do it quickly.
- **DO NOT** Touch an electrical system or circuit when your hands are wet.
- **DO NOT** use wires with poor insulation.
- **DO NOT** throw water on live electrical equipment in case of fire.
- **DO NOT** touch circuit with bare fingers.

ELECTRICAL SAFETY PRACTICES FOR CONTACTORS.



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SAFETY PRACTICES FOR CONTRACTORS

PLANNING:

- A single line diagram should be submitted by the contractor to safety department for approval before execution of work. Only after the approval of the single line diagram, the work should be executed.
- Only trained and authorized personnel should carry out any service or repair.

DISTRIBUTION PANELS:

- All electrical panels should be easily accessible & have a minimum statutory clearance.
- Ensure that panel's surrounding is free from combustible and flammable materials
- Tapping should be done from maintenance panel only after prior written approval.
- Prior to operation all power supply and protective devices should be checked.
- All panels should be duly grounded to avoid ground fault current.
- Panels should be fitted with Ground Fault Current Interrupters (GFCI) to avoid fault currents.
- When not in use, electrical panels doors should be closed and latched
- All distribution panels should display current rating.
- Panels should be at least to an IP44 rating to avoid dust entry and protect from wet condition.
- To avoid toppling, the base of the panels should be securely fixed to the ground.

ELECTRICITY SUPPLY FROM PANELS TO WORK AREA:

- Only designated person should be permitted to work on live electric circuit.
- Employees should be provided efficient personal protective equipment complying to BIS standards or any relevant international standards;

- Electrical contractors must ask themselves every day whether: -
 - 1) There are effective lockout/tagout (LOTO) & Permit to work procedures in place;
 - 2) Employees are properly trained for safe work practices;
 - 3) Safety talks are done with workers regularly;
 - 4) Portable electrical tools are grounded
 - 5) ELCB/RCD/Ground fault circuit interrupters and/or an assured equipment grounding program are in place for carrying out the assigned electrical work.
- Power should be tapped only from the designated sources as instructed.
- More attention to be given when power supplies serve more than one job, switching errors can result in energizing the wrong equipment, leading to possible hazards for the personnel.
- The panel board should be earthed.
- Provide signs or lights to warn personnel when the supply is energized.
- When using tools near electrical hazards, all tools must have a double insulated casing to prevent contact with energized parts.
- Before entering into power supply or associated equipment enclosure, following precautions should be taken: -
 - i. De-energisation of the equipment
 - ii. Open and lock out the main input circuit breaker, auxiliary power circuits should be checked
 - iii. Discard damaged cords, cords that become hot or cords with exposed wiring
 - iv. Power tools should be inspected for frayed power cords, broken plugs and cracked or broken housing. Electrical equipment should be Unplugged before repairing or servicing it
- Inspection, Testing and Labelling of the equipment must be ensured before start of the work.
- The post meant for cable support should be earthed duly
- Electrical cords should be placed away from areas where they may be pinched and areas where they may pose a tripping or fire hazard (e.g., doorways, walkways, under carpet etc.)
- Male and female industrial sockets should be used for connections of cords and no joints should be made with PVC tapes.
- Any crane or vehicle movement in the vicinity of supply lines should be monitored. Avoid any electrical contact with supply end and maintain sufficient clearance.

- Cable should run to the required length with minimum joints.
- No live ends of a conductor should be left unattended.
- Do not let Electrical equipment's to be suspended by the cords.
- Never leave holder without lamps or plugs.
- Ensure that apparatus is properly discharged before handling.
- All electrical cords and appliances must have apt weather protection.
- Do not attach Electrical cords to the building surfaces/structures.
- Caps should be used in unused sockets to avoid direct contact and mechanical damage.
- Ensure Firm Mounting of Receptacles in their enclosures. It should not move when the plug is inserted.
- Double insulation should be ensured in cables to prevent from fraying and possible mechanical damage.
- Cables running over the ground should be avoided if they pose a trip hazard or electrocution.
- Cables should be laid either underground or routed minimum 4m above from the ground level.
- The cable should be routed above the ground with the minimum height of 6m from the ground level where it is crossing the passage meant for vehicle movement, where underground laying is not possible.
- If the cables are laid underground, the cable route markings should be made with boards which should display the direction and the current rating.
- All extension boards should be minimum IP44 type.
- Use of extension cord should be minimized.
- When not in use, tools should be unplugged.
- Never whip a cord to unplug it.
- Ensure that the cable must be secured inside the equipment by non-conducting faces so that it cannot slip.
- Electrical equipment that has grounded three pronged plugs should only be used.
- If a prong breaks off inside an outlet. Do not attempt to remove by yourself.
- Do not remove the prongs of an electric plug. If plug prongs are missing, loose or bent replace the entire plug.

ADDITIONAL SAFE MEASURES:

- We should stay at least 10 feet away from overhead power lines.
- Ensure Proper illumination in all areas where electrical hazards are to be encountered. An emergency lighting system should be in place as well.
- Never drill a wall or floor without checking for any concealed utility electrical system.
- Specialized personnel protective equipment and clothing should be provided to the personnel to work on energized line or equipment.

The following safety features should be ensured while working with energized line/equipment:

- Insulated tools to avoid shocks.
- Rubber gloves.
- Shock resistant safety boots/shoes.
- Safety glasses/goggles.
- Flame resistant clothing if there exists any risk of an electric arc that could cause a fire.

Precautions to be taken while working with a hand tool:

- Follow permit system while doing maintenance/repair.
- Always wear hand gloves.
- Wear an electrical safety shoe while working with electricity.
- Stand on rubber mat while working.
- Residual current device (RCD) with 30mA sensitivity should be ensured in the operating circuit.
- Earth the phase conductors, while working and use locks to prevent inadvertent 'switching on' during repairs.
- Work on live terminal boxes, panels etc. should be done by using only one hand to the extent possible. Touching of any conducting/metal part with other hand should be avoided.
- Well insulated overall sheathed 3 or 4 core cables should be used.
- Joints in wires should be avoided.
- Do not use earth as neutral.

- Secure tight connections. A neutral disconnection stops the equipment but causes electrocution on touching the terminals (including the neutral terminal).
- The body of the equipment should be earthed using third core of the cable.

- Only flexible cables should be used to connect hand tools (avoid single strand /stiff wires). Use rubberised sheath / sleeves near terminals and ends, to avoid cuts due to frequent twisting/fatigue
- Follow proper interconnection techniques Plug a socket, terminal blocks, welding receptacles etc.
- Power supply for hand tools should be from centre tap earthed transformers.
- Use only flame proof equipment in classified areas.

SAFETY PRACTICES FOR CONTRACTORS

PLANNING:

- A single line diagram should be submitted by the contractor to safety department for approval before execution of work. Only after the approval of the single line diagram, the work should be executed.
- Only trained and authorized personnel should carry out any service or repair.

DISTRIBUTION PANELS:

- All electrical panels should be easily accessible & have a minimum statutory clearance.



Ensure that panel's surrounding is free from combustible and flammable materials



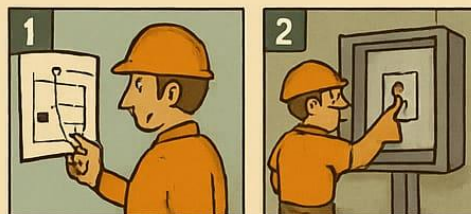
Check panels with current rating

Display current rating



Meeting Panel rating to avoid dust entry and protect from wet condition

To avoid fixing the electrical box to avoid dust entry and the ground



Only (independent) authorized personnel should check checklisting

ELECTRICITY SUPPLY FROM PANELS TO WORK AREA:

- Only designated person should be permitted to work on live electric circuit.
- Employees should be provided efficient personal protective equipment complying to BIS standards or any relevant international standards.
- Electrical contractors must ask themselves every day whether:
 - i. There are effective lockout/tagout (LOTO) & Permit to work procedures in place
 - ii. Employees are properly trained for safe work practices
 - iii. Safety talks are done with workers
 - iv. ELCR/RCD/Ground fault circuit interrupter (GFCI) to avoid fault currents



Employees should be provided efficient personal protective equipment complying to BIS standards or any relevant international standards:

- i. There are effective lockout/tagout (LOTO) & Permit to work procedures in place



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SAFETY AT HOME

SAFETY AT HOME



**Use ISI-marked
electrical devices**



**Turn off appliances
when not in use**



**Don't charge devices
under pillows
or with torn cables**



**Don't charge devices
under pillows or
with torn cables**

Common Electrical Hazards Around You



1

Water
near
electrical
outlets and
appliances



2

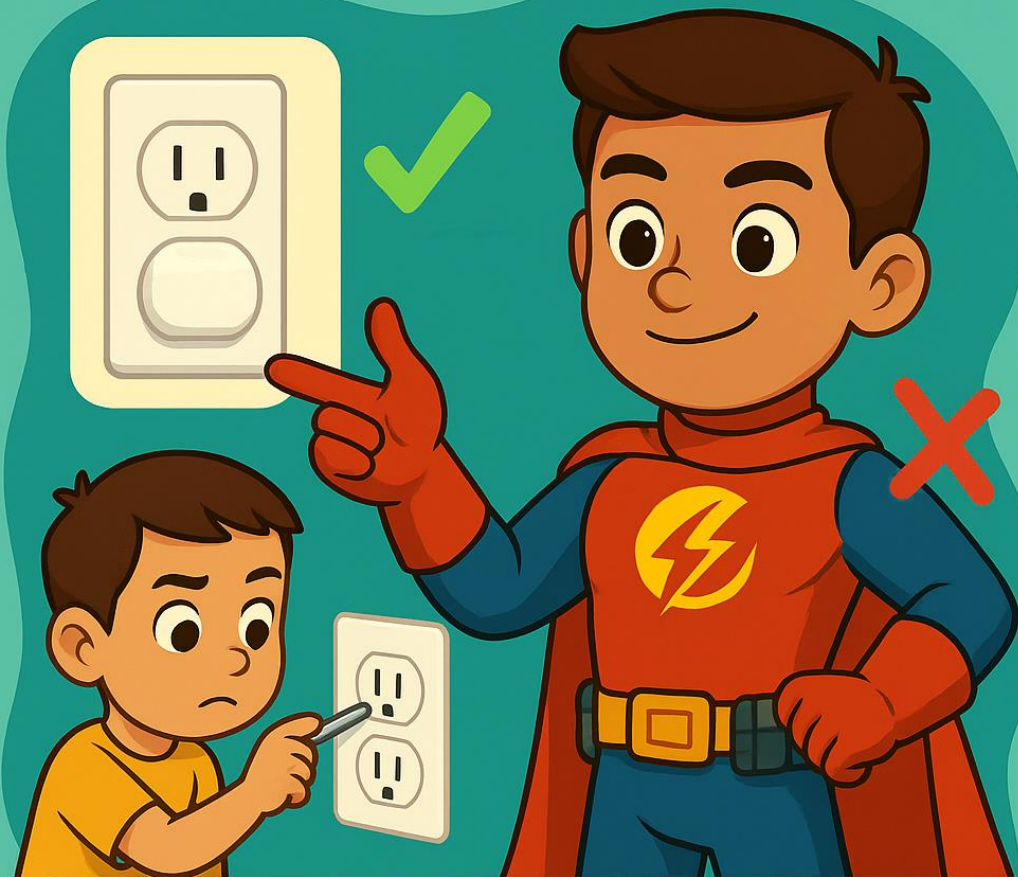
Plugging or
unplugging
cords unsafely



3

Climbing trees
or poles near
power lines

STAY SMART WITH SOCKETS!



- ✓ **Use safety covers**
- ✓ **Keep fingers and objects away**
- ✓ **Ask an adult for help**

COMMON ELECTRIC HAZARDS

FRAYED WIRES



Exposed copper
can cause electric shock

OVERLOADED SOCKETS



Can overheat
and catch fire

WET HANDS + SWITCHES



Shock hazard

CHEAP DEVICES



Can malfunction
or explode

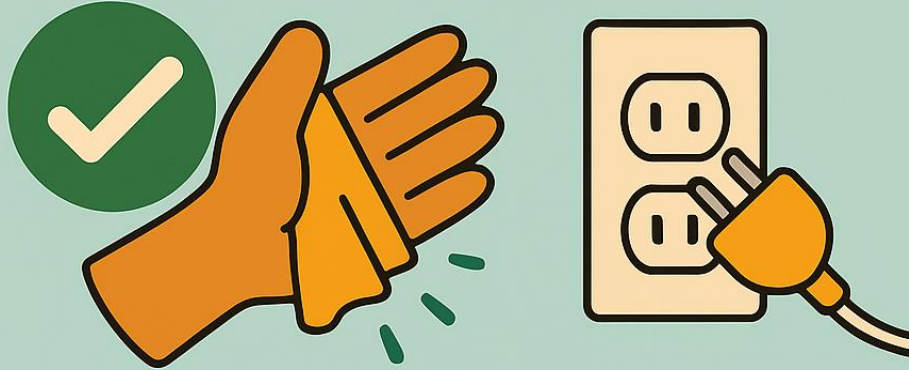
SMART CHARGING & DEVICES



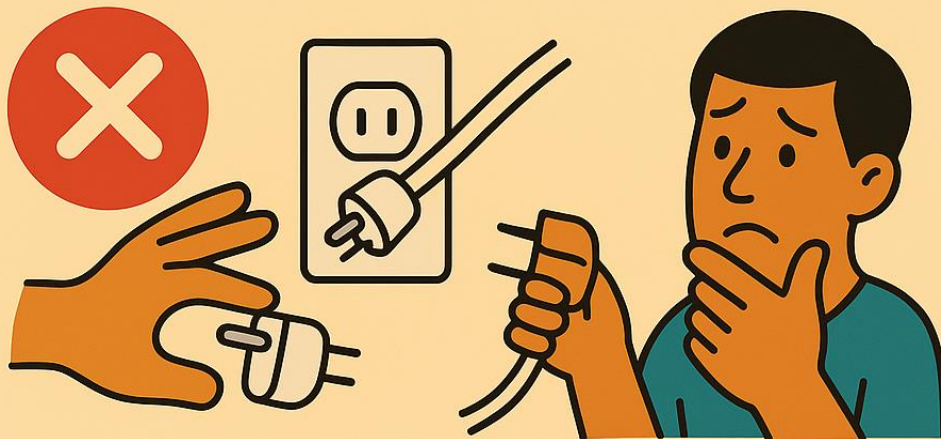
- ✓ Use BIS-approved chargers
- ✗ Don't overcharge devices overnight
- ✗ Avoid charging near water or soft bedding



- ✗ Avoid charging near soft bedding



**Use dry hands, proper plugs
and check wires**



**Don't touch metal parts,
yank cords, or fix issues alone**



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ELECTRICAL SAFETY NEAR / AT SCHOOL.

ELECTRICAL SAFETY NEAR/AT SCHOOL

- Schools should be located far away from the substations, supply lines.
- Open Distribution box, junction box, street box, etc. should not be there either in the school or on the way to school.
- Transformers should be fenced adequately.

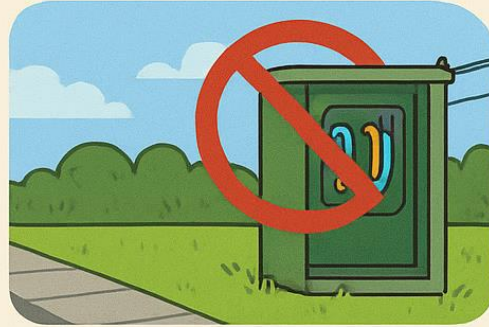
Do's

- DO ask for help when you need to use electrical equipment.
- DO have safety caps on all unused electrical outlets.

Don'ts

- DON'T ever climb the fence around an electrical substation.
- DON'T use electrical appliances near water.
- DON'T take shelter under any structure in rain/storm that has supply lines in its vicinity.
- Don't climb or touch a tree near power lines.

ELECTRICALL SAFETY NEAR/AT SCHOOL



Dos

- ✓ **DO** ask for help when you need to use electrical equipment
- ✓ **DO** have safety caps on all unused electrical outlets. 



Don'ts

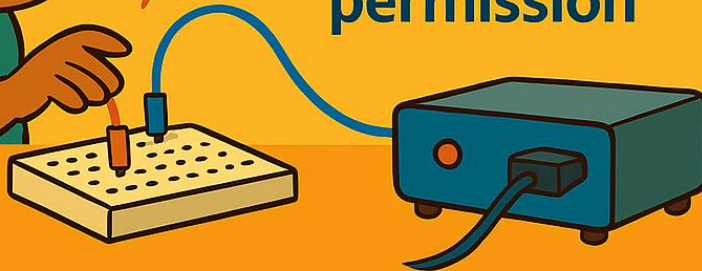
- ✗ **DON'T** ever climb the fence around an electrical substation
- ✗ **DON'T** use electrical appliances near water
- ✗ **DON'T** take shelter under any structure in rain/storm that has supply lines in it vicinity
- ✗ **DON'T** climb or touch a tree near power lines



SCHOOL & LAB SAFETY



**Never touch
wires or circuits
without
permission**



**Use low voltage
(12V max) for
school projects**



**Report damaged
sockets or plugs
to your teacher**





FOLLOW SAFE WORK PRACTICES WITH ELECTRICITY



Provide signs and barricades to warn of hazards

Provide signs and barricades to warn of hazards

GET ALL ELECTRICAL BEFORE THEY STU!



ROLE OF EARTHING IN ELECTRICAL SAFETY.

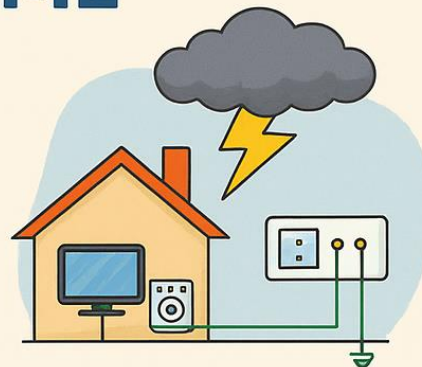
Earthing helps you & your home in following manners: -

- It ensures that all electrical equipment are at earth thus safeguarding from electric shocks through direct contact.
- It protects electrical equipment from getting damaged due to heavy currents.
- It maintains stable voltages in three phase circuits even under unbalanced load conditions.
- It protects tall buildings from getting harmed under lightning.

WHY EARTHING IS IMPORTANT FOR YOU & YOUR HOME

PROTECTION FROM ELECTRIC SHOCK

Ensures that all electrical equipment are at earth thus safeguarding from electric shocks through direct contact

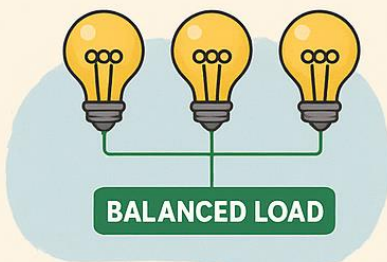


PROTECTS APPLIANCES FROM DAMAGE

Protects electrical equipment from getting damaged due to heavy currents

VOLTAGE STABILITY

Maintains stable voltages in three phase circuits even under unbalanced load conditions



PROTECTION FROM LIGHTNING

Protects tall buildings from getting harmed under lightning

THINGS THAT YOU SHOULD CARE

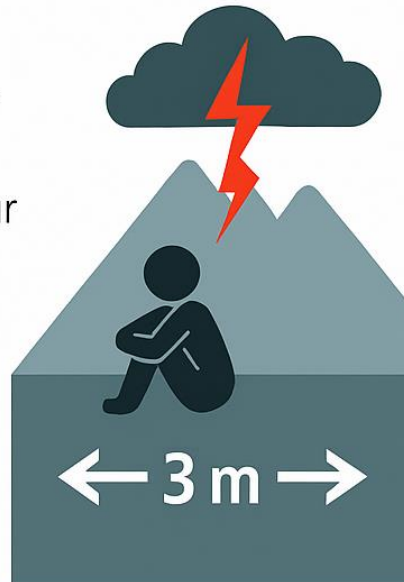
about while buying a new house

PRECAUTION AGAINST LIGHTNING.

Take following measures: in case inci lightning:

In Open Terrain:

- Put your feet as close together as possible, wrap your arms around your legs and tuck your head in. Keep a distance of 3 m to the next person.
- Avoid trees, groves, edges of the forest and wood poles of overhaud lines Keep a distance of 10 meter from all trees and limbs.



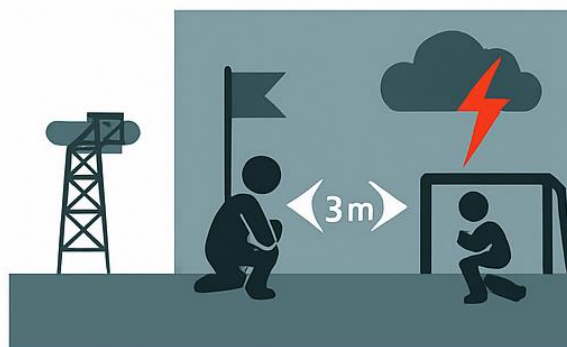
In the mountains:

- Keep off the peak, avoid being the highest point around.
- Keep a distance of at least 3 m from other people and metal objects such as hooks, ladders and ropes.



On a playing ground:

- Be at least 3 m away from radio masts and flagpoles.
- Put down flags, umbrellas and golf clubs.



On-board a boat

Go into the boat, crouch down and do not touch the rig or any other metal objects



Riding a bicycle



- If on a bicycle or motorcycle, stop riding, seek shelter (for example under a bridge) or crouch down at a distance of at least 3 meter from your vehicle

Inside a car

In case of heavy rain or rolls of thunder, it advisable to stop at the next possible place or wait until the thunderstorm has disappeared to avoid being blinded by the light and therefore losing control.



Indoors



Adequate lightning protection system shall be installed

LIGHTNING SAFETY GUIDELINES



During fishing or hunting:

- Lay down your fishing rod and seek shelter on land.
- Hunters should immediately leave open tree stands at the first sign of lightning and thunder.



During camping



- Keep a distance of at least 3 m from other tents and caravans. Crouch down in your tent on an insulating dry mat.
- Never erect your tent directly next to poles or pylons, at the edge of the forest or next to isolated trees.
- Use an insulating mattress and do not touch the tent poles during a thunderstorm

In open water:

- Leave the water right away and take cover at the first sign of lightning and thunder.

Under the Tree:

- Avoid trees, grooves and edges of the forest and keep a distance of 10 meters from any tree and limbs.



Avoid using the following during thunderstorm:



- Watching TV.
- Surfing the internet.
- Talking over mobile/telephone.
- Taking shower/bath.

What to do if someone is struck by lightning:

- Common injuries from being struck by lightning can be nerve and muscle paralysis, impaired eyesight and hearing as well as raised blood pressure.
- Try to calm the lightning victim if he/she is conscious. Softly talk to the victim until help arrives.
- If the victim is unconscious, put him/her in the recovery position. Check the victim's pulse and breathing. If the lightning victim has no pulse or heartbeat, lay him/her on his/her back and immediately begin cardiopulmonary resuscitation (heart massage and resuscitation). Do not stop until the victim starts to move, his/her chest begins to rise and fall or help arrives to take over.



Recovery position



Heart Massage



Resuscitation



LIGHTNING SAFETY

IN A BOAT



OFF A BICYCLE



INSIDE A CAR



INDOORS



WHAT TO AVOID DURING A THUNDERSTORM



WATCH TV

USE INTERNET

TALK ON PHONE

**IF SOMEONE IS
STRUCK BY LIGHTNING**



RECOVERY POSITION



HEART MASSAGE



HEART MASSAGE



CALM VICTIM



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SAFETY SIGN BOARDS & EQUIPMENTS

SAFETY SIGN BOARDS



Standard electrical warning sign – for use with – Live Bus-bar, Switchgear, live electrical equipment, warning of death, uninterruptible power supplies, working near live electrical equipment etc.



Example of electrical warning sign with mandatory instructions for example for use with multiple supplies, remotely/automatically controlled generating sets, main Intake Switch Sign, switch off electrical equipment before working etc.



Example of an electrical Prohibition Notice, primarily used as a Caution Notice



Electrical warning sign with mandatory action sign for locking HV enclosures, isolation prior to working for LV and HV equipment (Caution Notice), etc.



This area is under the control of
Estates Operations

In case of emergency or for general access, please
contact Estates Help Desk on extension 101, or
report to Estates Operations.

A Permit to Work may be required.



First Aid for High Voltage Electrical Shock Sign



Example of an electrical Prohibition Notice, primarily used as a Caution Notice.



Electrical Warning Sign used on electrical equipment such as switchgear and HV Enclosures.

Combination Signs



SAFETY SIGNS & THEIR MEANINGS

PROHIBITION SIGNS (DO NOT DO)

A sign prohibiting behavior likely to increase or cause danger.
eg. No Smoking
Colour : **Red**



MANDATORY SIGNS (MUST DO)

A sign prescribing specific behavior
eg. Hard hats must be worn.
Colour : **Blue**



SAFE CONDITION SIGNS (THE SAFE WAY)

A sign indicating emergency exits or first aid/rescue equipment.
Colour : **Green**



HAZARD SIGNS (CAUTION, BEWARE)

A sign giving warning of a hazard or danger.
eg. Danger Electricity
Colour : **Yellow**



FIRE SIGNS (FIRE EQUIPMENT)

A sign indicating the location of fire fighting equipment.
Colour : **Red**



INFORMATION SIGNS (GENERAL INFORMATION)

A sign providing general information.
eg. Toilets



SIGNAGES & DISPLAY BOARDS



PERSONAL PROTECTIVE EQUIPMENTS

HEAD PROTECTION



ULTRAVENT UDYOGI



LESLICO



SAVIOUR



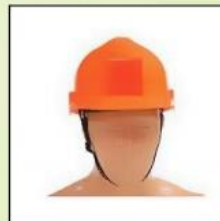
KARAM



LABOUR HELMET



BUMP CAP



EXECUTIVE HELMET



CONSTRUCTION HELMET

EYE PROTECTION



VISITOR



OVER SPECTACLE



CHEMICAL



WELDING



DUST



GRINDING



SMOKE



ARC WELDING

RESPIRATOR/MASK



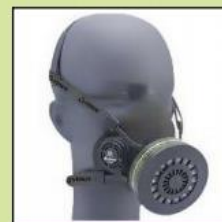
N95 RESPIRATOR



3 PLY MASK



REUSABLE HALF RESPIRATOR



SINGLE CARTRIDGE

EMERGENCY EVACUATION PLANS FOR HIGH RISE BUILDINGS



LONG DURABLE CORPORATE FRAME



MASTER PLAN

EMERGENCY LED LIGHTS



ANTISKID TAPES/AUTOGLLOW TAPES



DISPLAY BOARDS- FIRE & SAFETY



FIRE FIGHTING TRAINING BOARD



ELECTRICAL SAFETY BOARD



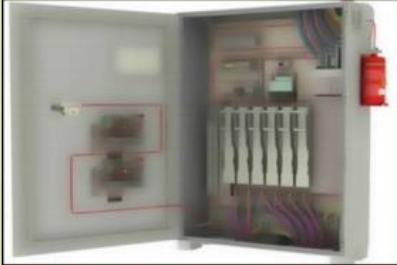
LOCKOUT STATION BOARD



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FIRE FIGHTING SYSTEMS



AUTOMATIC SUPPRESSION

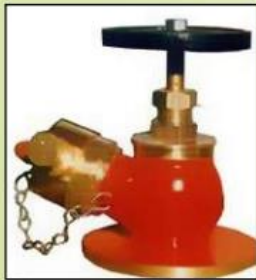


KITCHEN FIRE SUPPRESSION



FLOODING SYSTEM

HYDRANT PRODUCTS



HYDRANT VALVE



BRANCH PIPE



HYDRANT HOSE



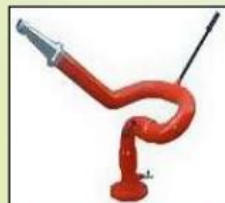
COUPLINGS



HOSE REEL DRUM



HOSE BOX



WATER MONITOR



FB5X



FAST ACTION NOZZLE



FOAM GENERATORS



MOBILE FOAM UNIT



AFFF SOLUTION



DIGITAL MPCB BOARD



SAFETY DISPLAY

FIRE EQUIPMENTS



ABC FIRE EXTINGUISHER



CO2 4.5KG FIRE EXTINGUISHER



MECHANICAL FOAM



KITCHEN FIRE EXTINGUISHER



MODULAR FIRE EXTINGUISHER



TROLLEY FIRE EXTINGUISHER



WATER TYPE



FIRE EXTINGUISHER REFILLING



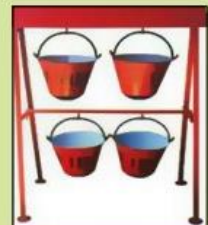
FIRE MAN AXE



FLOOR STAND



FIRE EXT. TROLLEY



FIRE BUCKET

HANDGLOVES



PU COATED



CUT RESISTANT



LATEX COATED



POLKA DOTTED



EXAMINATION GLOVES



CHEMICAL GLOVES



ELECTRICAL GLOVES



RUBBER GLOVES



HEAT RESISTANT



WELDING



ALUMINIZED



STEEL MESH



IMPACT SAFE



HAND SLEEVE



CRYOGENIC GLOVES



FIRE MAN



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CHEMICAL HANDLING & STORAGE PRODUCTS



PORTABLE SPILL KIT



TROLLEY SPILL KIT



SORBENT PADS



CLEAN ROOM STICKY MATS



PIG CONTAINMENT DECK



DRUM PALLET



SPILL BLOCKER DIKE



SPILL TRAY



SAFETY CAN



PLUNGER CAN



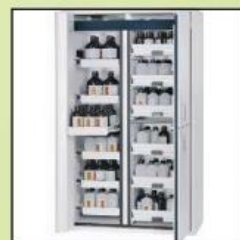
DISTRIBUTION CAN



FM APPROVED CABINETS



TYPE 90 CABINETS



FLAAMMABLE LIQUID STORAGE



AUTOCLOG SHOES



AUTOCLOG GOGGLES



PALM PALTE



RESTROOM MATS



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BLOWER



ALUMINIUM LADDER



MULTI GAS DETECTOR



OXYGEN KIT

ELECTRICIAL SAFETY PRODUCTS



INSULATING MAT



ELECTRICAL GLOVES



ARC FLASH SUIT



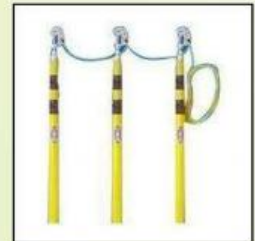
RESCUE HOOK



ELECTRICAL FACESHIELD



DIELECTRIC BOOTS



DISCHARGE ROD



CABLE LOCATOR/GPRS SURVEY



INSULATED TOOLS



FRP LADDER

FALL PROTECTION



ECO 1



TANGO 2



FALL ARRESTER BLOCK



ROPE GRAB ARRESTER



ENERGY ABSORBER



LIFELINE



ANCHORAGE PLATE



POLYMIDE ROPE



VERTICLE LIFE LINESYSTEM



RESCUE KIT



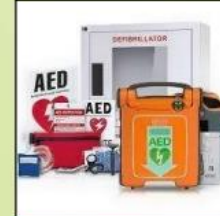
EMERGENCY EQUIPMENTS



BURN FREE DRESSING



FIRST AID BOX



AED



AMBU BAG

EMERGENCY EQUIPMENTS



PPE CABINET



BREATHING APPARATUS



FIRE MAN SUIT



SMOKE HOOD



MEGAPHONE



STRETCHER



WHEEL CHAIR



EVACUATION CHAIR



HELMET MOUNTED



INTRINSICALLY SAFE TORCH



SEARCH LIGHT



REMOTE AREA LIGHT SYSTEM

MISCELLANEOUS PRODUCTS



EMERGENCY SIREN



WIND SOCK



FIRE BLANKET



EYEWASH BOTTLE



PORTABLE EYEWASH STATION



EYEWASH FOUNTAIN/SHOWER



SNAKE CATCHER

BODY SUIT / COVERALL



DUST



CHEMICAL



PVC SUIT



APRON



ALUMINIZED SUIT



NOMEX SUIT



ARC FLASH SUIT



BOILER SUIT

SAFETY SHOE



FS213



UD16



CHETAK



POWER



GALAXY



FS05



ENGINEER



PROTECTA



BROWN



COFFER



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TRAFFIC SAFETY PRODUCTS



REFLECTIVE JACKET



TRAFFIC CONE



SPRING POST



LED BATON



CORNER GUARDS



SPEED BREAKER



RUBBER WHEEL STOPPER



METRO BARRIERS



PLASTIC BARRIER



CONVEX MIRROR



UVSM



SPARK ARRESTER



WHEEL LOCK



QUEUE MANAGER



BOOM BARRIER



DFMD



METAL DETECTOR



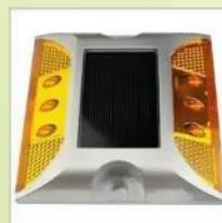
SECURITY FENCE BARRIER SYSTEM



ANTISTATIC PAINT



THERMOPLASTIC PAINT



SOLAR ROAD STUDS



SITE SAFETY



The following approved PPE must be worn on this site at all times



Safety helmets



Eye protection



High visibility clothing



Protective gloves



Protective footwear



All visitors and delivery drivers must report to the site office



SITE SAFETY

Under the Health and Safety at work work Act 1974 All persons entering this site must comply with all regulations under this act. All visitors must report to the site office and obtain permission to proceed on the site or any work area. Safety signs and procedures must be observed and personal protection and safety equipment must be used at all time.



Construction Work in progress.

Parents are advised to warn children of the dangers of entering of this site.



Unauthorised entry to this site is strictly forbidden.



Safety helmets must be worn.



Protective footwear must be worn.



High Visibility clothing must be worn.



Eye protection must be worn.



This is a no smoking site.

No Hat No Boot No Job!



**FIRST AID IS AVAILABLE
FROM THE SITE OFFICE.**



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FIRST AID

First Aid:

- A person getting electric shock needs to be disconnected from the source of electrical power. The disconnecting switch/breaker should be identified and turned off.
- Unfortunately, if the connected device cannot be located, the victim can be pried or pulled from the circuit by an insulated object such as a dry wood board, piece of non-metallic conduit, or rubber electrical cord.
- Immediate medical response should be provided to victims i.e. check for breathing and pulse, then apply Cardio-Pulmonary Resuscitation (CPR) as necessary to maintain oxygenation.
- If a victim is still conscious, he needs to be closely monitored and cared for until trained emergency response personnel arrive.
- The victim should be kept warm and comfortable to avoid danger of physiological shock.

FIRST AID: ELECTRIC SHOCK



A person getting electric shock needs to be disconnected from the source of electrical power



The victim can be pried or pulled from the circuit by an insulated object such as a dry wood board



Immediate medical response should be provided to victims



The victim should be kept warm and comfortable

Immediate medical response should be provided to victims
ie check for breathing and pulse

The victim should be kept warm and comfortable

ELECTRIC SHOCK & ITS MANAGEMENT

Regional Ud 3-2AB Joint Group

CAUSES



Contact with
a live electric wire



2 Give the burnt area
with cold water

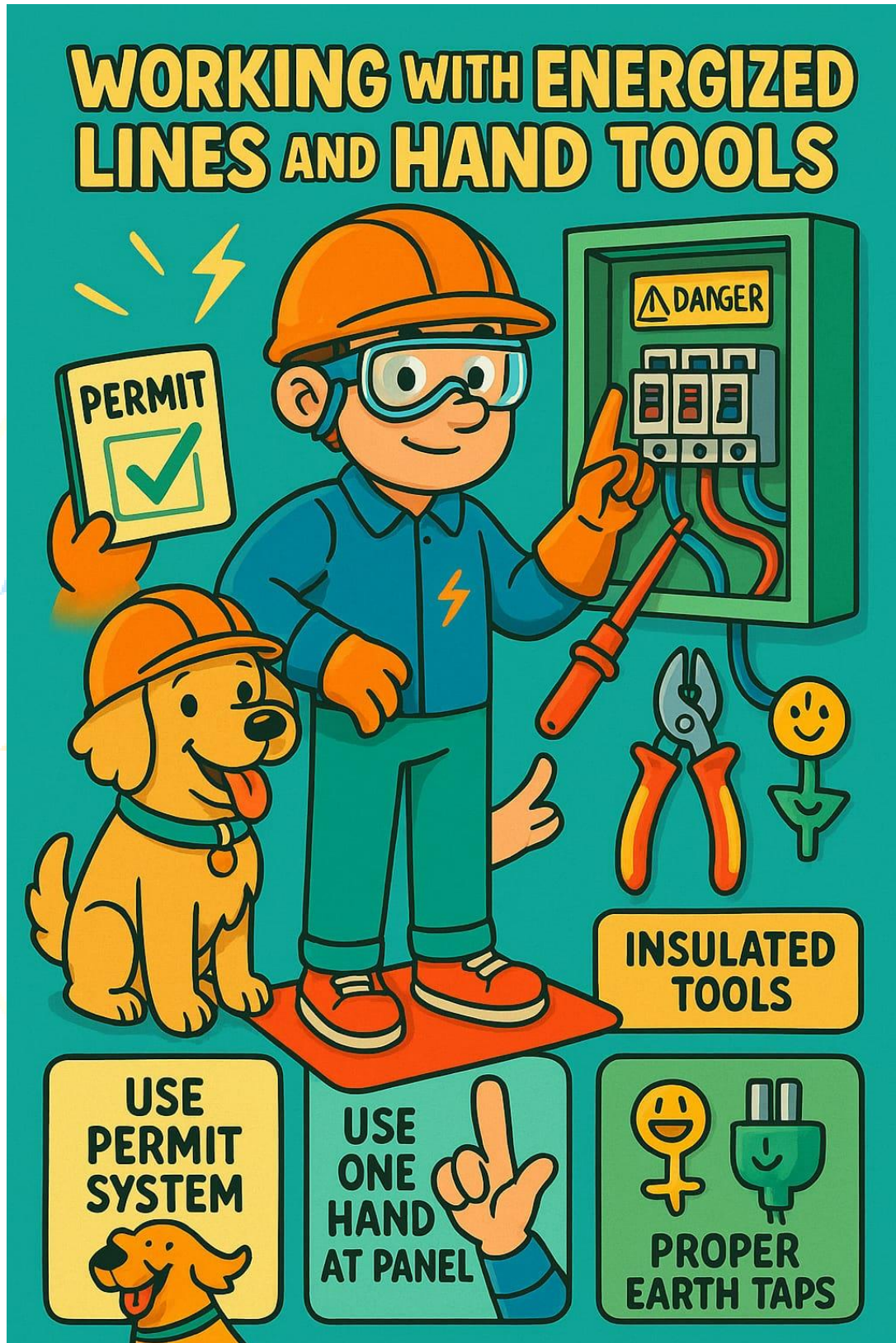
MANAGEMENT



1 Stand on dry insulated
material like wood,
folded newspaper,
rubber mat, etc.

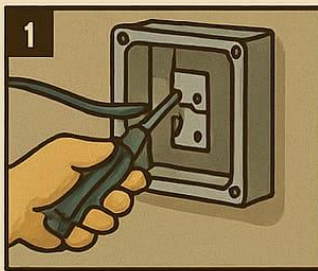


2 Give the burnt area
with cold water

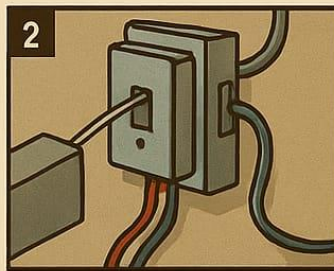


HOW TO DEAL WITH SHORT CIRCUITS

CAUSES OF SHORT CIRCUITS



1
Faulty circuit wire insulation caused due to old or damaged insulation



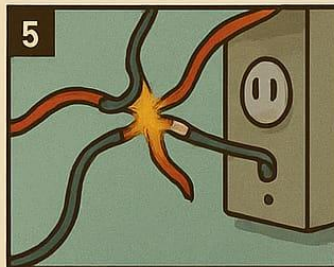
2
Overloading single socket outlet by using multiple pin plugs



3
Use of undersized wire not enough to handle the estimated electrical load on the circuit



4
Loose wire connections sometime allows neutral and live wire to touch which can cause



5
Faulty appliance wiring can occur in the plugs, in the power cords or inside the device itself



5
Locate the tripped circuit breaker

DEALING WITH SHORT CIRCUITS



1
Disconnect the power source before taking up electrical repair work



2
Unplug appliances before trying to inspect or repair them



3
Call Competent / Certified Electrician to Repair wiring problem

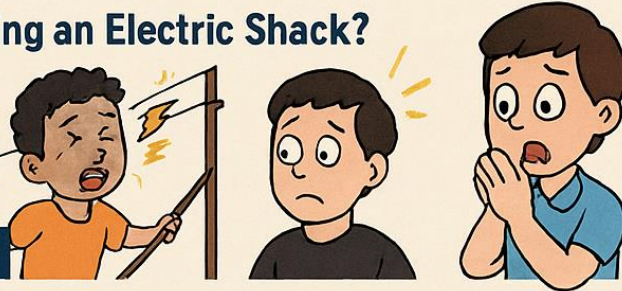
ELECTRIC SHOCK & ITS MANAGEMENT

Regional UD 3-2AB Joint Grup

What Happens During an Electric Shock?

Contact with a high-tension live wire can be deadly.

Immediate and correct action is critical.



DAY 3 – CASE EXAMPLE

EMERGENCY MANAGEMENT – STEP BY STEP



- Do not touch the person directly.
- Stand on dry insulated materials like wood
- Folded newspaper
- Rubber mat, etc.

2. BURNS & LIQUID CONTACT

- Pour cool clean water over burns.
- Covering with clean, dry cloth or sterile bandage.
- Do not apply oil, butter, or cream.
- Do not give person anything to drink



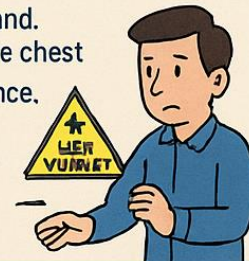
3. IF THE PERSON IS UNCONSCIOUS

- Check if the person is unconscious.
- Always wear rubber gloves if available,
- Use insulated tools immediately.



4. CHEST COMPRESSIONS (CPR)

- Use the heel of hand in the center of the chest
- Keep a safe distance.
- Wait for power to be cut off by red switch.



5. LOW VOLTAGE SAFETY PRECAUTIONS



- Always wear dry rubber gloves if available.
- Use insulated tools only.
- Call for professional help immediately.

7. AFTER THE INCIDENT

- Call a doctor or ambulance immediately.
- Take casualty to nearest hospital.
- Keep them calm still.
- Monitor breathing and pulse until help arrives



Note to Students: Always be aware of your surroundings. Stay safe around electricity

ELECTRIC SHOCK EMERGENCY RESUSCITATION

Safeguard Yourself

If a patient collapses due to ELECTRIC SHOCK, break the circuit by switching off the current at the mains, removing the plug or wrenching the cable free. If unable to reach the cable, socket or mains, use or stand on some DRY non-conducting material and using a broom, safety hook, wooden chair or stool, push the casualty away from the source of electricity



Check Casualty Response

Once the area is safe

If unconscious, open the airway and check for normal breathing (for up to 10) seconds.



If the casualty is breathing, place in recovery position, dial 102 for an ambulance, monitor and record breathing every ten minutes or until medical assistance arrives.

If Casualty is Not Breathing Normally

If the casualty is not breathing normally, dial 102 immediately and commence CPR

Commence CPR

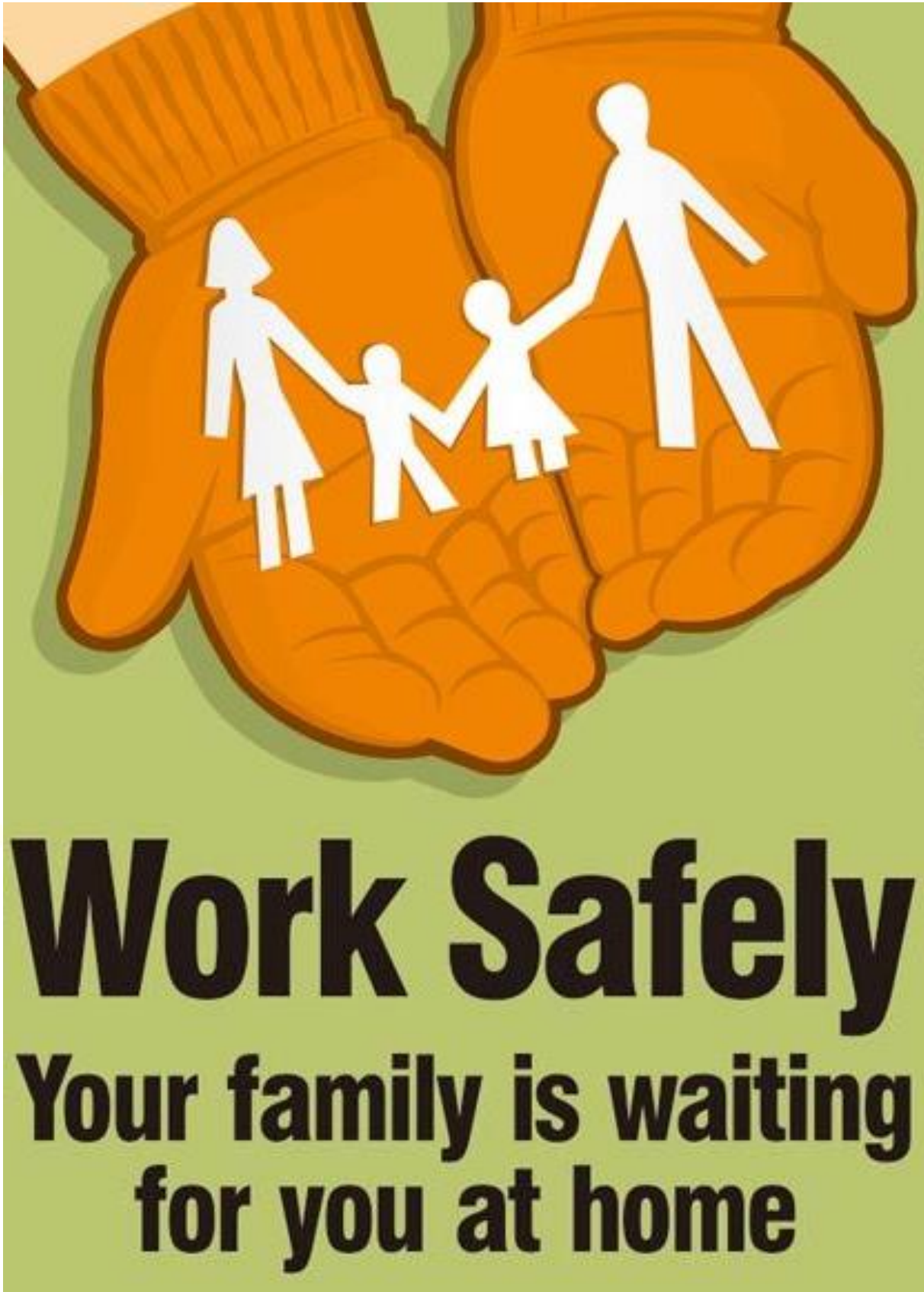
Commence CPR, Alternate 30 chest compressions with two rescue breaths.



Repeat this sequence as necessary



FOR CHILDRENS



REFERENCES: -

-GOOGLE

-USED AI

- DEEPSEAK AI
- CHAT GPT

-YOUTUBE (viewers are requested to watch these videos for better understanding.)

- <https://youtu.be/2N7fOYZYx2g?si=dgf7lhdt-N7J6TJ6>
- https://youtu.be/h8IeJZsz-Sw?si=BaFQ_Pqfq4QHKMJg
- <https://youtu.be/ggJo6m8NZtA?si=F6Mg8KCd5g5ceArE>
- <https://youtu.be/Kt75TTtrxA?si=72k2eI2WXzeeBCTD>
- <https://youtube.com/shorts/6fxwFEUmWi8?si=R4LuovbE7wuQq1Od>
- <https://youtu.be/OPs4MQOIZAQ?si=l9KgSRDARpVh1K-6>
- <https://youtu.be/BYWU-gEB7nI?si=yRBa3s9g3mlOafjR>
- https://youtu.be/qcj10SScYMY?si=ca_opD8OppYabnCl

NOTE – Some of the animated posters were generated by using AI, there might be some spellings & Grammer errors. So, please read the theory thoroughly, and use images and posters to understand the situation and imagine.